



Your home, perfectly in sync

NestSync is a smart roommate management system that simplifies shared living by combining expense tracking, chore management, and communication in one platform. It helps roommates stay organized, transparent, and in sync—reducing conflicts and making co-living effortless.

Proposed By: Group 9

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# 1 Project Description

## **Project Overview**

NestSync is a full-stack Roommate Management System designed to simplify shared living by integrating expense management, chore coordination, shared scheduling, and communication into a single platform. The system enables roommates to manage finances, responsibilities, and household activities efficiently, reducing conflicts and improving transparency.

## **Motivation**

As students and working professionals often live with roommates, managing shared expenses, chores, and communication can become complex and lead to misunderstandings. Existing solutions like Splitwise focus only on financial tracking and lack holistic household management features. NestSync addresses this gap by providing an all-in-one solution for co-living coordination.

## **Goals**

- Provide a centralized platform for managing all roommate-related activities.
- Ensure fairness, transparency, and accountability in shared living environments.

## **Scope & Feasibility**

The system will be developed as a web application using modern technologies (e.g., React, Spring Boot, MySQL). The MVP will include expense tracking, chore management, and notifications, with additional features like shared calendar and maintenance tracking in advanced iterations.

# 2 Requirements

## Target Users

- Students living in shared apartments
- Working professionals with roommates
- Flatmates in shared housing environments

## Functional Requirements

- The system shall allow users to create and join a roommate group.
- The system shall allow users to add, edit, and delete shared expenses.
- The system shall calculate and display individual balances for each user.
- The system shall generate optimized settlement suggestions between roommates.
- The system shall allow users to assign and manage household chores.
- The system shall send notifications for pending payments and assigned chores.
- The system shall allow users to maintain a shared grocery list.
- The system shall allow users to log and track maintenance issues.

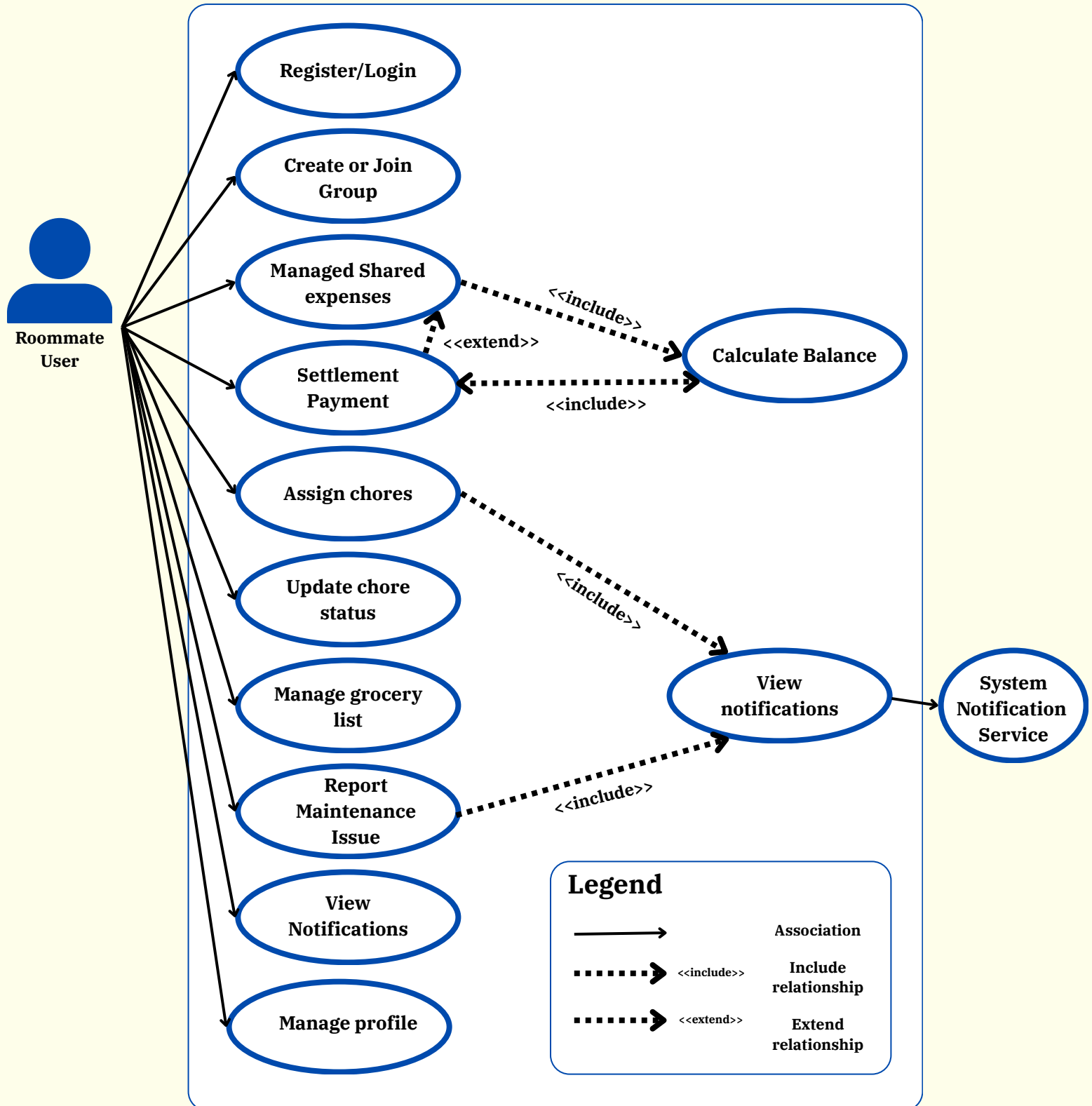
## Non-Functional Requirements

- Usability: The system shall provide an intuitive and user-friendly interface.
- Performance: The system shall respond to user actions within 2 seconds.
- Security: The system shall ensure secure authentication and data protection.
- Scalability: The system shall support multiple groups and concurrent users.
- Reliability: The system shall ensure data consistency for financial transactions.
- Availability: The system shall be accessible 24/7 with minimal downtime.

# 3 Glossary

| Term              | Definition                                         |
|-------------------|----------------------------------------------------|
| User              | A registered individual using the system           |
| Group             | A collection of roommates sharing a living space   |
| Expense           | A financial transaction shared among roommates     |
| Balance           | Amount owed or to be received by a user            |
| Settlement        | Payment made to clear outstanding balances         |
| Chore             | Household task assigned to a user                  |
| Notification      | Alert sent to users regarding actions or reminders |
| Maintenance Issue | A problem reported in the household (e.g., repair) |

# 4 Use Case Diagram



# 5 Use Case Descriptions

## Use Case 1: Add Shared Expense

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>            | Roommate User                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Description</b>      | A roommate user can add a shared household expense and split it among selected roommates.                                                                                                                                                                                                                                                                                                                                                         |
| <b>Preconditions</b>    | User must be logged in and must belong to a roommate group.                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Postconditions</b>   | The expense is saved, and each roommate's balance is updated.                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Priority</b>         | Essential                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Frequency of Use</b> | Frequent                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Flow of Events</b>   | <ol style="list-style-type: none"><li>1. User selects Add Expense.</li><li>2. User enters expense title, amount, date, category, and notes.</li><li>3. User selects roommates included in the expense.</li><li>4. User chooses split type: equal, percentage, or custom.</li><li>5. System calculates each roommate's share.</li><li>6. User reviews and submits the expense.</li><li>7. System saves the expense and updates balances.</li></ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"><li>1. If required fields are missing, system shows an error message.</li><li>2. If percentage split does not total 100%, system asks the user to correct it.</li><li>3. User updates the information and submits again.</li></ol>                                                                                                                                                                              |

# 5 Use Case Descriptions

## Use Case 2: Settle Balance

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>            | Roommate User                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description</b>      | A roommate user can view outstanding balances and mark a payment as settled.                                                                                                                                                                                                                                                                                                                                             |
| <b>Preconditions</b>    | User must be logged in and must have an active balance in the group.                                                                                                                                                                                                                                                                                                                                                     |
| <b>Postconditions</b>   | The settlement is recorded and group balances are updated.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Priority</b>         | Essential                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Frequency of Use</b> | Frequent                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Flow of Events</b>   | <ol style="list-style-type: none"><li>1. User opens the Balances page.</li><li>2. System displays who owes whom and how much.</li><li>3. User selects Settle Payment.</li><li>4. System shows optimized payment suggestions.</li><li>5. User selects a settlement and confirms payment.</li><li>6. System records the settlement.</li><li>7. System updates balances and sends notification to involved users.</li></ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"><li>1. If user cancels settlement, no balance changes are made.</li><li>2. If payment confirmation is missing, system keeps the balance pending.</li></ol>                                                                                                                                                                                                                             |

# 5 Use Case Descriptions

## Use Case 3: Manage Chores

|                         |                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>            | Roommate User                                                                                                                                                                                                                                                                                                                                                      |
| <b>Description</b>      | A roommate user can create, assign, and track household chores.                                                                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>    | User must be logged in and must belong to a roommate group.                                                                                                                                                                                                                                                                                                        |
| <b>Postconditions</b>   | Chore is created or updated, and assigned roommate receives notification.                                                                                                                                                                                                                                                                                          |
| <b>Priority</b>         | Essential                                                                                                                                                                                                                                                                                                                                                          |
| <b>Frequency of Use</b> | Frequent                                                                                                                                                                                                                                                                                                                                                           |
| <b>Flow of Events</b>   | <ol style="list-style-type: none"><li>1. User opens the Chores page.</li><li>2. User selects Create Chore.</li><li>3. User enters chore name, due date, recurrence, and assignee.</li><li>4. System saves the chore.</li><li>5. Assigned roommate views the chore.</li><li>6. Roommate marks chore as completed.</li><li>7. System updates chore status.</li></ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"><li>1. If no roommate is selected, system asks user to choose an assignee.</li><li>2. If chore is overdue, system sends a reminder notification.</li></ol>                                                                                                                                                                       |

# 5 Use Case Descriptions

## Use Case 4: Report Maintenance Issue

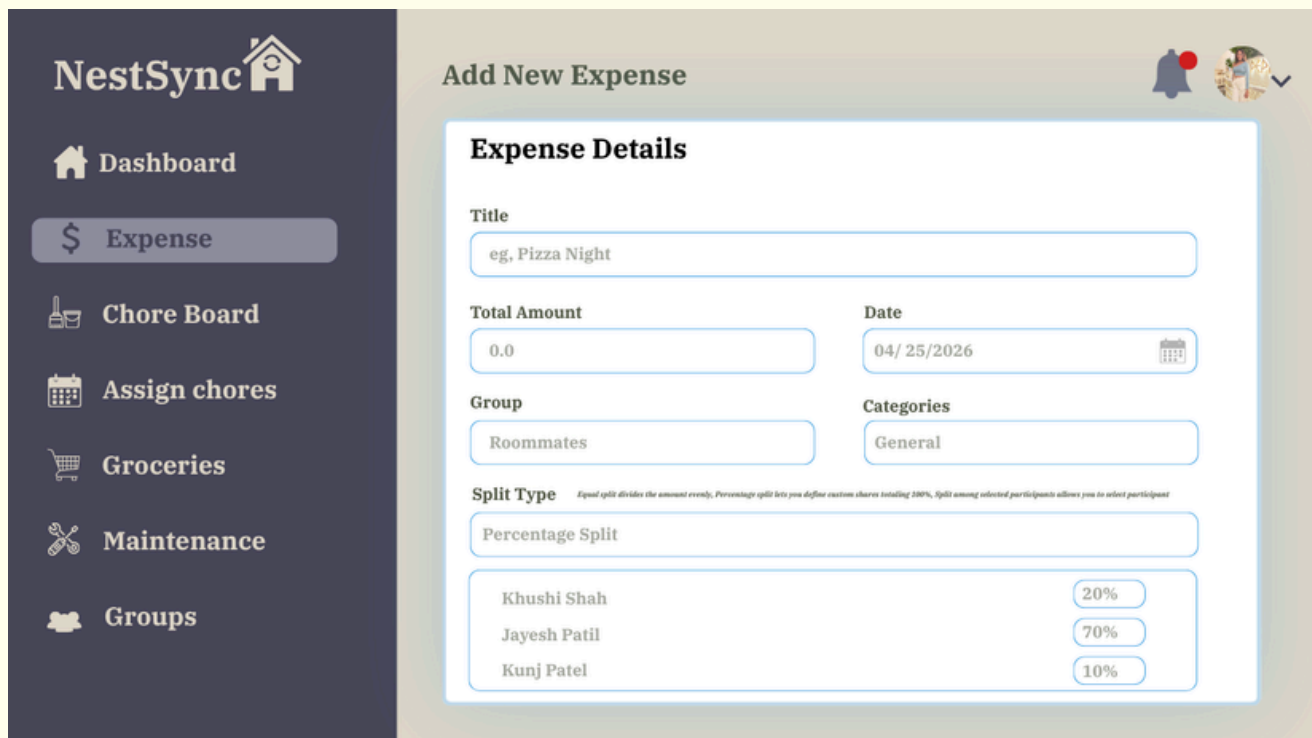
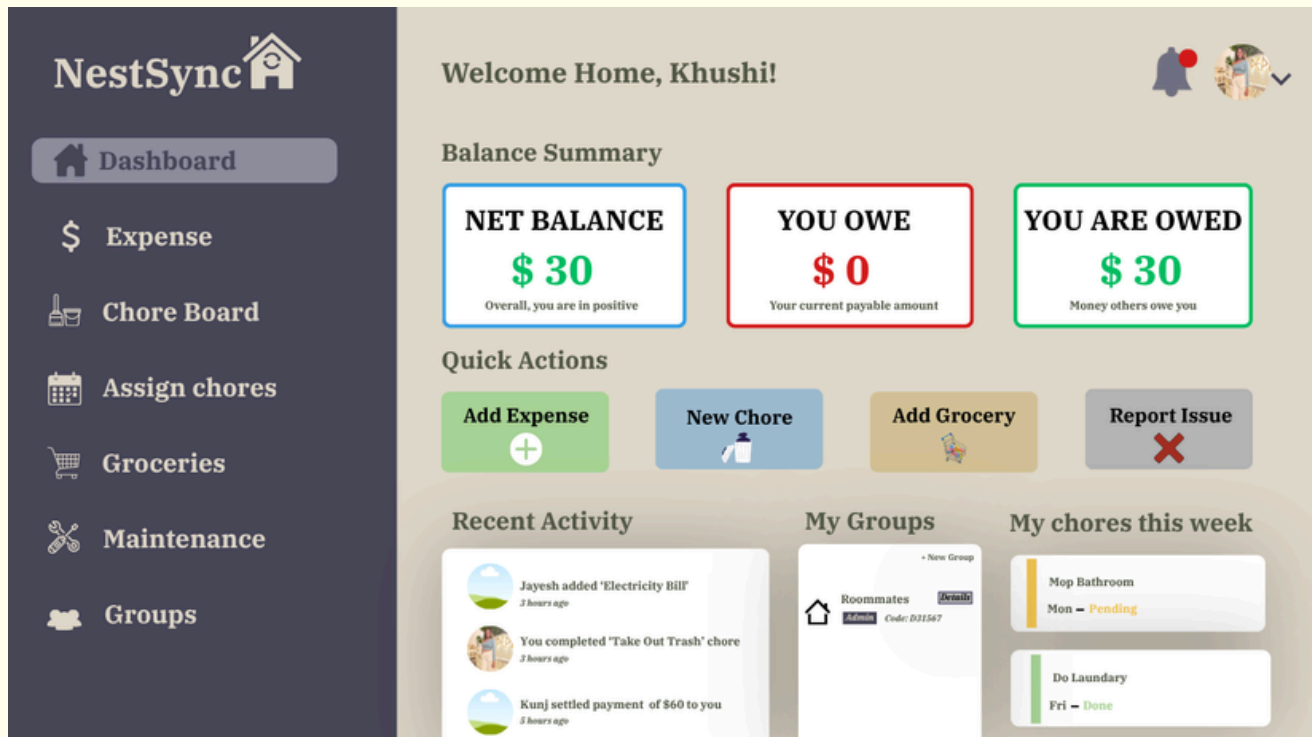
|                         |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>            | Roommate User                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Description</b>      | A roommate user can report a household maintenance issue and track its status.                                                                                                                                                                                                                                                                                                                                     |
| <b>Preconditions</b>    | User must be logged in and must belong to a roommate group.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Postconditions</b>   | Maintenance issue is recorded and visible to all group members.                                                                                                                                                                                                                                                                                                                                                    |
| <b>Priority</b>         | Important                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Frequency of Use</b> | Occasional                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Flow of Events</b>   | <ol style="list-style-type: none"><li>1. User opens the Maintenance page.</li><li>2. User selects Report Issue.</li><li>3. User enters issue title, description, priority, and optional image.</li><li>4. User submits the issue.</li><li>5. System saves the issue as Open.</li><li>6. Roommates can view and update issue status.</li><li>7. System displays status as Open, In Progress, or Resolved.</li></ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"><li>1. If issue description is missing, system asks user to complete required fields.</li><li>2. If issue is resolved, user marks it as completed.</li></ol>                                                                                                                                                                                                                     |

# 5 Use Case Descriptions

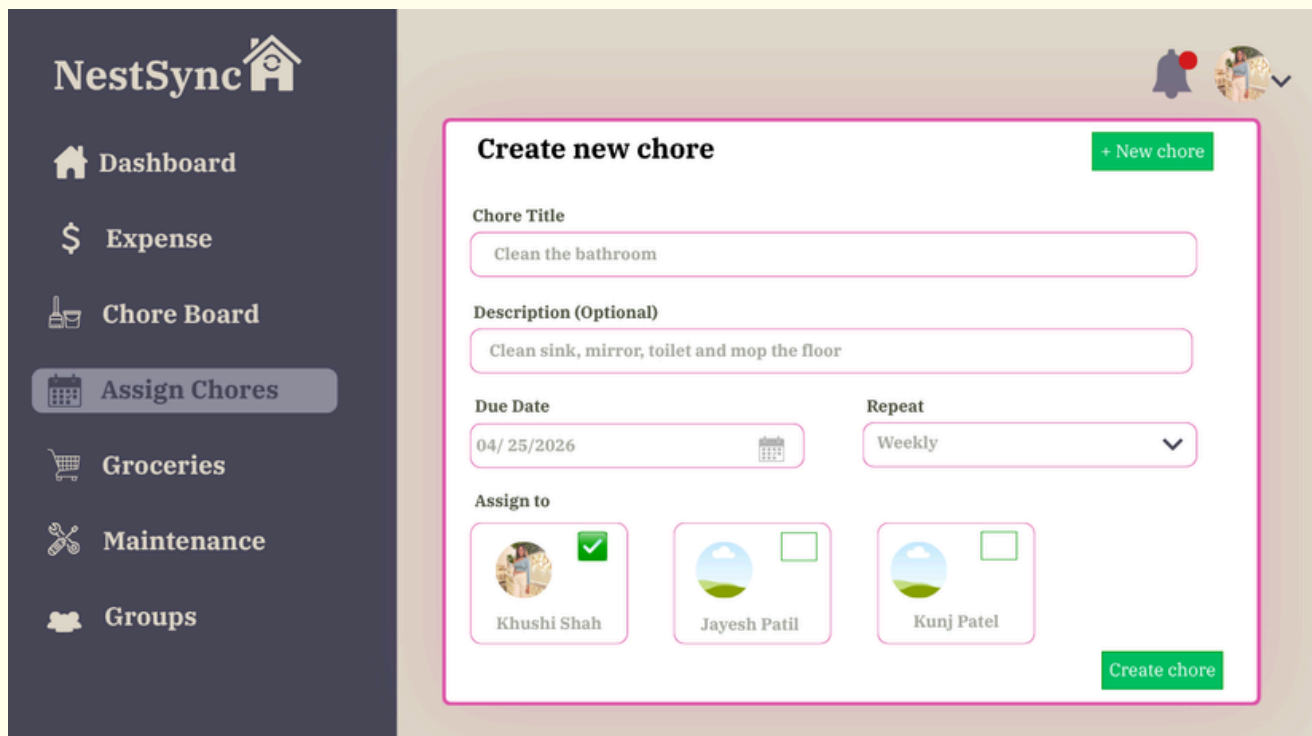
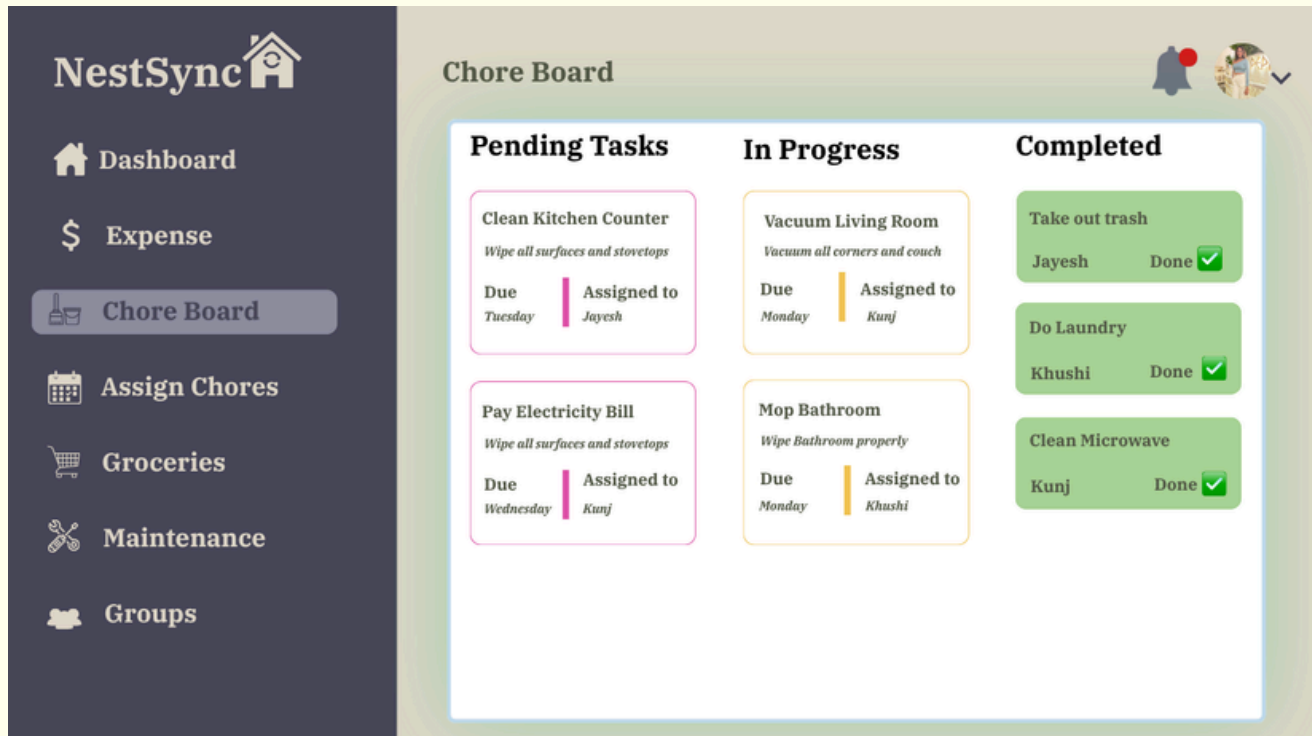
## Use Case 5: View Notifications

|                         |                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actor</b>            | Roommate User                                                                                                                                                                                                                                                                                                                                          |
| <b>Description</b>      | A roommate user can view reminders related to payments, chores, and maintenance updates.                                                                                                                                                                                                                                                               |
| <b>Preconditions</b>    | User must be logged in.                                                                                                                                                                                                                                                                                                                                |
| <b>Postconditions</b>   | User views latest notifications and may take required action.                                                                                                                                                                                                                                                                                          |
| <b>Priority</b>         | Important                                                                                                                                                                                                                                                                                                                                              |
| <b>Frequency of Use</b> | Frequent                                                                                                                                                                                                                                                                                                                                               |
| <b>Flow of Events</b>   | <ol style="list-style-type: none"><li>1. User opens the Notifications panel.</li><li>2. System displays payment, chore, and issue reminders.</li><li>3. User selects a notification.</li><li>4. System redirects user to the related page.</li><li>5. User completes required action.</li><li>6. System updates notification status as read.</li></ol> |
| <b>Alternative Flow</b> | <ol style="list-style-type: none"><li>1. If there are no notifications, system displays "No new notifications."</li></ol>                                                                                                                                                                                                                              |

# UI Mockups



# 6 UI Mockups



# 6 UI Mockups

NestSync

 Dashboard

 Expense

 Chore Board

 Assign Chores

 Groceries

 Maintenance

 Groups

## Grocery List

All Items

My Items

Purchased

 8  
Total Items

 7  
To Buy

 1  
Purchased

Categories

 Produce (5)

| Product  | Quantity | Added by |
|----------|----------|----------|
| Bananas  | 6 pcs    | Khushi   |
| Spinach  | 1 bunch  | Kunj     |
| Tomatoes | 10 pcs   | Jayesh   |
| Lemons   | 4 pcs    | Khushi   |

 Dairy & Eggs (3)

|               |           |        |
|---------------|-----------|--------|
| Milk          | 2 gallons | Khushi |
| Eggs          | 12 pcs    | Jayesh |
| Greek Yoghurt | 1 tub     | Kunj   |

Add Item

Product Title

eg. Bananas

Categories

Produce

Quantity

6 pcs

Group

Roommate

Member

Khushi

+ Add Item

NestSync

 Dashboard

 Expense

 Chore Board

 Assign Chores

 Groceries

 Maintenance

 Groups

## Maintenance

 6  
Total Requests

 2  
Resolved

 1  
Open

Maintenance Requests

 Kitchen Sink Leaking  
Water is leaking from the pipe under the sink.  
Added by Khushi  
April 1, 2026 Plumbing

 AC Not Cooling  
The living room AC is on but not cooling properly.  
Added by Jayesh  
April 15, 2026 Electric

 Bathroom Light Flickering  
The light keeps flickering when it's on.  
Added by Kunj  
April 20, 2026 Electric

Issue Title

eg. Kitchen Sink Leaking

Categories

Plumbing

Description (Optional)

Please describe the issue in detail

Date

April 1, 2026

Member

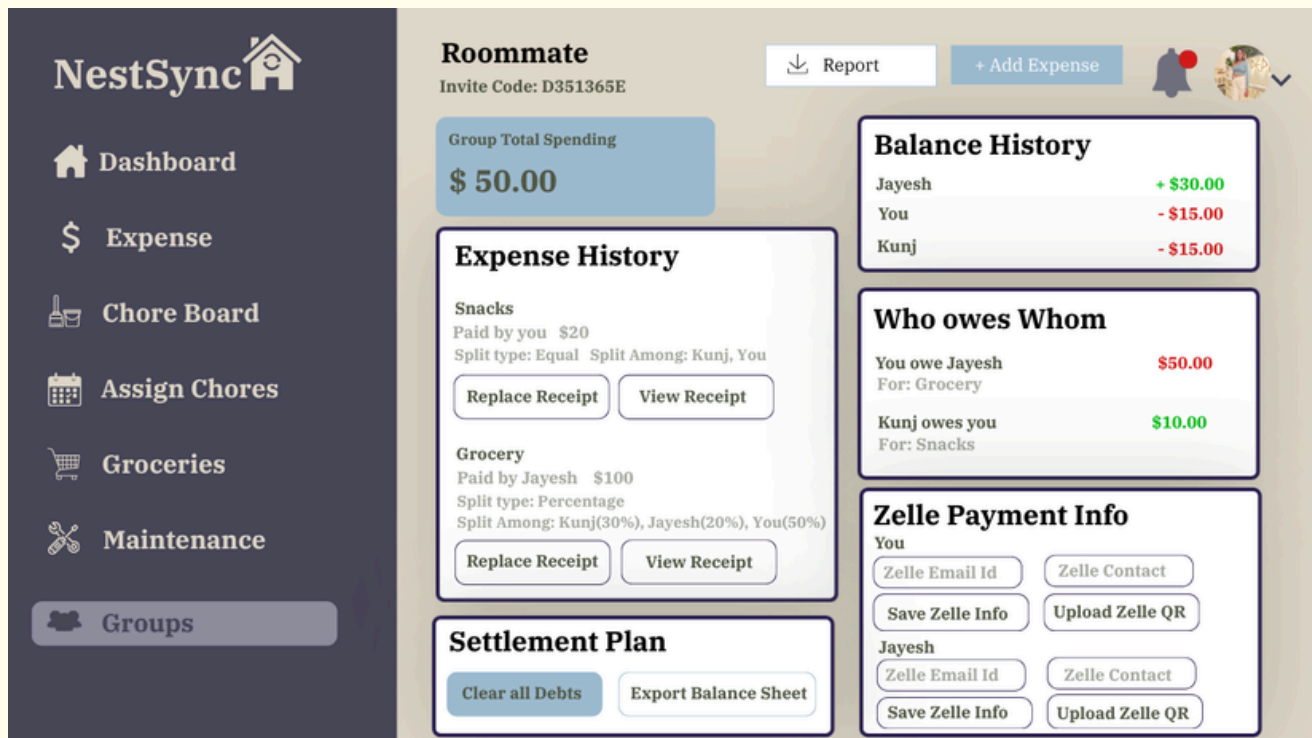
Khushi

Add photos (Optional)

Drag & Drop or click to upload (PNG or JPG)

+ Add Issues

# UI Mockups



## HCI Principles:

- Simple navigation
- Consistent layout
- Clear visual hierarchy
- Minimal cognitive load